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Economic Report and Production Analysis for Brazil

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0. Executive Summary

In this report, we aim to uncover the economic path Brazil went through during the period from 2009 to 2014. Specifically, we dive into the topic through a macroeconomic overview of Brazil from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We find that during the period of our interest, Brazil demonstrated a moderate level of economic growth with an annual GDP growth rate of 8.1%, concurrent with a decrease in economic complexity as compared with other major emerging economies.

The level of economic progress during the period, we believe, can be partially explained through analysis of the economic and product complexity structure of Brazil on the industry group level. The product classification system we adopt allow us to measure the actual development for various industry groups and to observe that Brazil aggressively invested in developing daily instrument, high-tech and machinery industries during the period, while its emerging peers and developed economies around the globe have adopted paths that are more well-suited to their current stances in the product space.

Based on our analysis of the opportunity gain and a sensitivity analysis of major industry groups, Brazil could further its gains by prioritizing stone, service, and clothing industry groups. We further note that Brazil has an ill-matched development strategy that indicates a particularly low level of structural optimality on the complexity level, having an adjusted R-square of merely 0.8%. Since a significant positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, Brazil indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale, and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.

1. Report Overview

In this report, we aim to uncover the economic path Brazil went through during the period from 2009 to 2014. Specifically, we dive into the topic through a macroeconomic overview of Brazil from the perspective of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect countries' production capability and production structure, respectively. We then analyze the industrial framework and changes of Brazil during the period based on quantitative measures such as Product Complexity Index (PCI) and Product Family Complexity Index (PfCI), followed by a detailed analysis of how such changes have matched with development strategies as quantitatively measured by opportunity gains on an industry-level. We conclude our report with macro-level results and predictions of Brazil, as deduced from regression models we have established on a global scale, which takes into account countries' stances in their product space.

2. Macroeconomic Results

2.1 GDP Ranking and Changes

As the 8th largest economy in the world by GDP, Brazil has a GDP of 1.7trillion USD in 2009 and 2.5trillion USD in 2014, exhibiting an extraordinary level of economic output. Figure 1 illustrates the time-series GDP results for majoremergingeconomies from 1996-2016, and Brazildemonstrates a high level of consistency in terms of both GDP and GDP growth.However, its growth rate is not significant as compared with many other emerging economies in the world, withan annual growth of 8.1% during the period from 2009 to 2014.This number is slightlybelow the annual growth rate of the other five leading emerging economies, which have a geometric average growth rate of 10.2% and an arithmetic average growth rate of 10.5%. The contrast is particularly significant with China, one of the leading emerging economy in the world, which has an average annual growth rate of 15.5% during the same period. The pace of economic growth in Brazil, we believe, can be reflected through changes in ECI and PCI, and can be further explained by a suboptimal set of production strategies adopted by Brazilfrom 2009 to 2014, which we will expand at the later stage of the report.

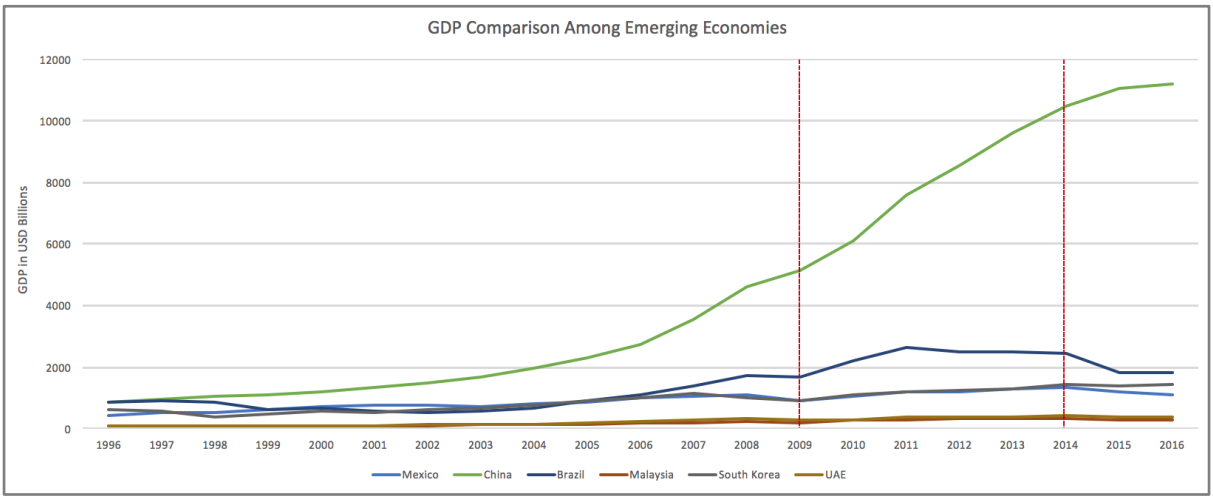


Figure 1: GDP Comparison Among Major Emerging Economies

2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), as defined by Ricardo Hausmann and Cesar Hidalgo in 2009 to explain the knowledge accumulated in a country's population, exhibits similar pattern for Brazil. As illustratedinFigure 2 below, Brazil exhibits a slight decrease in terms of its production capability from 2009 to 2014, as measured by ECI. This is in direct contrast to countries such as China and South Korea, which demonstrate an advantage in terms of both the ECI and GDP indexes.

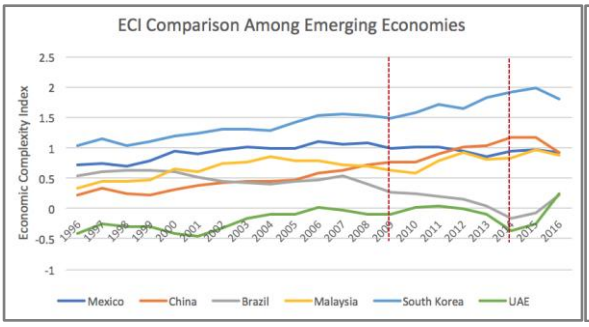


Figure 2: ECI Comparison

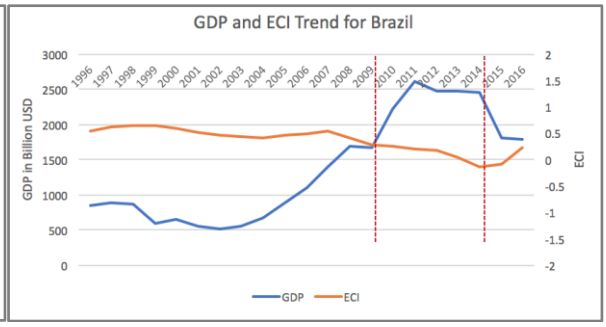


Figure 3: GDP and ECI (Brazil)

Figure 3 offers a further comparison of GDP and ECI over the entire time span from 1996 to 2016 for Brazil. The stagnancy in GDP during the period from 2009 to 2014 is accompanied by a stagnancy in ECI, indicting the potential that the mild economic growth experienced by Brazil during the period from 2009 to 2014 might be attributed to its slow-growing ECI, or the ineffectiveness of the country in developing a competitive product space that helps to accumulate complicated knowledge and technology.In order to provide a more concrete answer to why this has happened, we will dive into an analysis of Brazil's production structure and its product space designs in the next few chapters.

3. ProductionAnalysis

3.1 Product Categorization

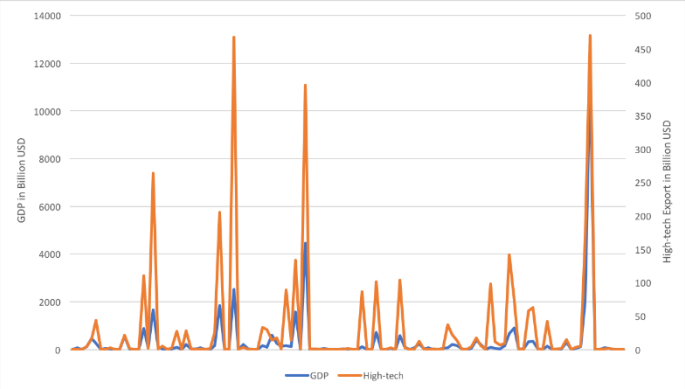
Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Product in our data source are labelled from 010000-999999, with the first two-digits representing a general category to which the product belongs. Based on the 99 categories as extracted from the label, we further group the products into 18 most widely-accepted industries, as summarized in Table 1.

Note that the high-tech industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technical gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), high-tech industry has the highest PfCI among all industry groups.

Category	Description
Animal and By-product	All kinds of meat, livestock and related by-products
Vegetable, Fruit and Plant	All kinds of plants, edible vegetables and fruits
Food and Preparation	Edible preparations, raw cooking materials and appliances
Mineral and Stone	Mining industry for mineral and stone materials
Chemical	Raw chemical materials and by-products
Chemical Manufacturing	Advanced chemical products and pharmaceutical products
Raw Manufacturing	Basic products made from plastic, rubber, leather, etc.
Wood Product	Book, paper and other elementary products related to wood
Clothing and Accessory	All kinds of fabrics, filament, clothing and accessories
Textile	All kinds of textile articles and related products
Metal	Mining of all sorts of raw metal
Machinery	Machinery and mechanical appliances, including electrical equipment
Transportation	Vehicles and locomotives including aircraft, ship, railway, etc.
Weapon	Explosives, arms and ammunition
Daily Instrument	Basic daily instruments such as sport requisites, timing and measuring tools
Precious Stone	Refined or processed stone products such as pearl and ceramics
Service	Financial, business, legal, communication, travel services, etc.
High-tech	Agglomeration of advanced products with highest PCIs

Table 1: Categorization of Products

We further observe a strong positivecorrelation between the GDP and the export of high-tech products. As shown by Figure 4 below, a high adjusted R-square of 67.6% exists between GDP and high-tech exports, indicating that more successful economics, as measure by GDP, tend to be those that are capable of producing and exporting high-tech products as we define them.



Statistics	Value
Multiple R	0.824
R-Square	0.679
Adjusted R^2	0.676
Variable	**Value**
Coefficient	12.20
t-stat	15.61
P-value	0.000(***)

Figure 4: Regression of GDP and High-tech Exports

3.2 Actual Development

Based on the product categorization aforementioned, we aim to capture Brazil's actual production strategy between 2009 and 2014 on a complexity level.In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is in developing products within that group. Defining this net distance change as the Actual Developmental Index (ADI), our export data allows us to calculate and compare ADIofBrazilamong the 18 product groups in 2009 and 2014 through subtraction, where a

positive ADI indicates a shortening of the distance (i.e., advancement of the industry), and a negative ADI indicates the reverse. As ranked in Table 2, during the period from 2009to 2014, Brazil has focused on products with a relatively high level of complexity and those requiring advancedskills, including daily instrument, high-tech products and those related to machinery, which are unusual for a developing economy.

Category	ADI	Category	ADI
1. Daily Instrument	0.157	2. High-tech	0.156
3. Machinery	0.125	4. Metal	0.094
5. Chemical Manufacturing	0.088	6. Weapon	0.087
7. Textile	0.083	8. Service	0.065
9. Wood Product	0.063	10. Transportation	0.059
11. Animal and By-product	0.057	12. Mineral and Stone	0.056
13. Food and Preparation	0.046	14. Raw Manufacturing	0.018
15. Chemicals	0.009	16. Vegetable, Fruit and Plant	0.004
17. Precious Stone	0.003	18. Clothing and Accessory	0.001

Table 2: Actual Developmental Index (ADI) of Brazil

3.3 Regional and Global Comparison

Table 3and 4 below summarizes how the production strategy adopted by Brazil compares with other emerging economics and a selected group of leading economies in the world, as measured in ADI. It can be directly observed that Brazil focused on products with a higher complexity level, when compared against its emerging counterparts such as Mexico and even China, which give a considerable level of attention to more basic product groups such as those related to food production.

Ranking	Brazil	Mexico	China	India
1	Daily Instrument	Vegetable, Fruit and Plant	Precious Stone	Daily Instrument
2	High-tech	Animal and By-product	Service	High-tech
3	Machinery	Chemicals	High-tech	Wood Product
4	Metal	Mineral and Stone	Machinery	Machinery
5	Chemical Manufacturing	Metal	Chemicals	Chemicals

Table 3: ADIComparison on a Regional Scale

Ranking	USA	Germany	Singapore	France
1	Service	Chemicals	Machinery	Raw Manufacturing
2	Chemical Manufacturing	Service	Transportation	Mineral and Stone
3	Food and Preparation	High-tech	Daily Instrument	Precious Stone
4	Textile	Transportation	Chemicals	Chemical Manufacturing
5	Clothing and Accessory	Food and Prep	High-tech	Chemicals

Table 4: ADI Comparison on a Global Scale

Table 4 further examines Brazil's development strategy against those of more developed economies on a global scale. Reasonably, developed economics such as USA, Germany, Singapore and Japan have devoted the majority of their efforts into highly complicated industry groups while emerging economies such as Brazil still partially relies on intermediate industry groups like metal and daily instrument. This trend holds true in general, regardless of the continent or the time period, and serves as an indicator for which phase of development

a country is currently in as evaluated from their production strategy. Based on this standard, Brazil is indeed adopting a moderate but well-timed strategy that seeks development in industry groups with an intermediate/high level of complexity.

4. Product Space Design

4.1 Suggested Development Vector

As aforementioned, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined inHausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP=0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter in the product as compared to global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in term of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP=1) has an opportunity gain of zero. And unreasonable investment on certain product p would lead to a relative decrement in export of more profitable products, and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economics strength, those having higher ECI, or in other words, more reasonable economics structure isoften in better shape.

Ranking	Brazil	Mexico	China	India
1	Precious Stone	Precious Stone	Machinery	High-tech
2	Service	Daily Instrument	High-tech	Machinery
3	Clothing and Accessory	High-tech	Chemicals	Daily Instrument
4	Daily Instrument	Machinery	Daily Instrument	Transportation
5	High-tech	Mineral and Stone	Textile	Chemical Manufacturing
6	Chemical Manufacturing	Vegetable, Fruit and Plant	Metal	Weapon
7	Machinery	Food and Preparation	Raw Manufacturing	Wood Product
8	Textile	Transportation	Wood Product	Metal
9	Chemicals	Animal and By-product	Clothing and Accessory	Raw Manufacturing
10	Metal	Clothing and Accessory	Service	Animal and By-product
11	Raw Manufacturing	Chemical Manufacturing	Chemical Manufacturing	Food and Preparation
12	Wood Product	Weapon	Precious Stone	Mineral and Stone
13	Weapon	Chemicals	Weapon	Precious Stone
14	Animal and By-product	Service	Animal and By-product	Vegetable, Fruit and Plant
15	Mineral and Stone	Wood Product	Mineral and Stone	Clothing and Accessory
16	Vegetable, Fruit and Plant	Textile	Vegetable, Fruit and Plant	Chemicals
17	Food and Preparation	Metal	Food and Preparation	Service
18	Transportation	Raw Manufacturing	Transportation	Textile
ECI	0.266	1.083	0.754	0.196

Table 5: Opportunity Gain Ranking on a Comparable Scale

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 and 6 summarize how the SDV ofBrazil compares with a selected group of comparable and leading economies in the world, with the industry groups ranked in decreasing order of significance.Dark grey indicates positive opportunity gain, gray indicates zero opportunity gains, and light grey indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic transition, those countries merely need to maintain a steady and balanced economic development. In contrast, developing countries have nearly half of industries with positive opportunity gain, and share common interests in industrial

manufacturing and high-tech industries. It can be concluded that Brazilshares the characters of developing countries, with abalanced economics structure that leans towards intermediate-levelindustries.

Ranking	USA	Germany	Singapore	France
1	Machinery	Machinery	Machinery	High-tech
2	High-tech	High-tech	High-tech	Machinery
3	Transportation	Transportation	Chemicals	Daily Instrument
4	Chemicals	Metal	Mineral and Stone	Chemicals
5	Textile	Raw Manufacturing	Transportation	Textile
6	Clothing and Accessory	Daily Instrument	Weapon	Weapon
7	Metal	Chemical Manufacturing	Chemical Manufacturing	Raw Manufacturing
8	Raw Manufacturing	Wood Product	Daily Instrument	Wood Product
9	Daily Instrument	Weapon	Textile	Clothing and Accessory
10	Chemical Manufacturing	Service	Metal	Metal
11	Weapon	Chemicals	Raw Manufacturing	Transportation
12	Wood Product	Clothing and Accessory	Precious Stone	Chemical Manufacturing
13	Service	Mineral and Stone	Wood Product	Animal and By-product
14	Vegetable, Fruit and Plant	Animal and By-product	Clothing and Accessory	Food and Preparation
15	Precious Stone	Precious Stone	Animal and By-product	Service
16	Animal and By-product	Textile	Vegetable, Fruit and Plant	Vegetable, Fruit and Plant
17	Food and Preparation	Food and Preparation	Service	Precious Stone
18	Mineral and Stone	Vegetable, Fruit and Plant	Food and Preparation	Mineral and Stone
ECI	1.533	1.917	1.578	1.465

Table 6: Opportunity Gain Ranking on a Global Scale

4.2 Sequential Development Strategy

The sequential economic development strategy aims to approximate the most efficient order of industry development.As discussed in the previous section, opportunity gain measures the potential profit in terms of economic complexity when a country invests on a product p to an extent such that the MCP bumps up from 0 to 1 (i.e., the country becomes a significant exporter in that product). Obviously, the difficulty of bumping up MCP varies across industries. Therefore, we capture this variation of difficulty by defining the concept of marginal opportunity gains.

$$marginal\ opportunity\ gains_{cp} = \frac{opportunity\ gains_{cp}}{\left(\frac{minimum\ \ increment\ \ of\ export\ \ of\ product\ \ p\ for\ country\ \ c}{present\ \ export\ \ of\ product\ \ p\ for\ country\ \ c}\right)}$$

In each iteration, we order all 18 industries based on their current marginal opportunity gain, and prioritize the industry with the highest marginal opportunity gain. We increase the prioritized industry's export to an extent that the MCP firmly bumps up to 1.Following the routine above, we derive the iterative strategy as shown in Table 7, ranked by marginal opportunity gains in decreasing order. Dark grey indicates positive opportunity gain, gray indicates zero opportunity gains, and light grey indicates negative opportunity gains.

4.3 Strategy Interpretation

Table 7 shows that at present Brazil lacks necessary industrial power, and therefore barely has advantage in manufactured goods. Basic products such as precious stone, textile and chemical manufacturing are the foundation of intermediate-level products, and industries such as daily instrument and service are the second easiest step. As a result, gaining comparative edges in these industries will rocket the ECI of Brazil, and it is noteworthy that the significance of high-tech drops extensively after the development of servicere-opens the door of light industries and intermediate-level industries such as transportation, chemicals and manufacturing. Because Brazil has a large gap between its current product space and advanced product groups such as high-tech and machinery, it is extremely costly for Brazil to

develop these two products even in the final stable stage, and Brazil may have a compelling industrial structure without a comparative edge in these two industry groups.

Before Iteration	Iteration 1	Iteration 2	Iteration 3
Precious Stone	Textile	Clothing and Accessory	Chemical Manufacturing
Textile	Clothing and Accessory	Chemical Manufacturing	Daily Instrument
Clothing and Accessory	Chemical Manufacturing	Daily Instrument	Service
Chemical Manufacturing	Daily Instrument	High-tech	High-tech
Daily Instrument	High-tech	Service	Machinery
High-tech	Machinery	Machinery	Animal and By-product
Machinery	Service	Animal and By-product	Vegetable, Fruit and Plant
Service	Animal and By-product	Vegetable, Fruit and Plant	Food and Preparation
Animal and By-product	Vegetable, Fruit and Plant	Food and Preparation	Mineral and Stone
Vegetable, Fruit and Plant	Food and Preparation	Mineral and Stone	Chemicals
Food and Preparation	Mineral and Stone	Chemicals	Raw Manufacturing
Mineral and Stone	Chemicals	Raw Manufacturing	Wood Product
Chemicals	Raw Manufacturing	Wood Product	Clothing and Accessory
Raw Manufacturing	Wood Product	Textile	Textile
Wood Product	Metal	Metal	Metal
Metal	Transportation	Transportation	Transportation
Transportation	Weapon	Weapon	Weapon
Weapon	Precious Stone	Precious Stone	Precious Stone

Iteration 4	Iteration 5	Iteration 6	Iteration 7
Daily Instrument	Service	Textile	Transportation
Service	Animal and By-product	Transportation	Precious Stone
High-tech	Vegetable, Fruit and Plant	Daily Instrument	Clothing and Accessory
Animal and By-product	Food and Preparation	Precious Stone	Daily Instrument
Vegetable, Fruit and Plant	Mineral and Stone	Clothing and Accessory	Chemical Manufacturing
Food and Preparation	Chemicals	Chemical Manufacturing	Service
Mineral and Stone	Chemical Manufacturing	Service	Machinery
Chemicals	Raw Manufacturing	Machinery	High-tech
Chemical Manufacturing	Wood Product	High-tech	Animal and By-product
Raw Manufacturing	Clothing and Accessory	Animal and By-product	Vegetable, Fruit and Plant
Wood Product	Textile	Vegetable, Fruit and Plant	Food and Preparation
Clothing and Accessory	Metal	Food and Preparation	Mineral and Stone
Textile	Transportation	Mineral and Stone	Chemicals
Metal	Weapon	Chemicals	Raw Manufacturing
Transportation	Daily Instrument	Raw Manufacturing	Wood Product
Weapon	Precious Stone	Wood Product	Textile
Precious Stone	Machinery	Metal	Metal
Machinery	High-tech	Weapon	Weapon

Iteration 8	Iteration 9	Iteration 10	Iteration 11
Chemicals	Clothing and Accessory	Precious Stone	Chemical Manufacturing
Clothing and Accessory	Precious Stone	Chemical Manufacturing	Daily Instrument
Precious Stone	Daily Instrument	Daily Instrument	Service
Daily Instrument	Chemical Manufacturing	Service	High-tech
Chemical Manufacturing	Service	High-tech	Machinery
Service	High-tech	Machinery	Animal and By-product
High-tech	Machinery	Animal and By-product	Vegetable, Fruit and Plant
Machinery	Animal and By-product	Vegetable, Fruit and Plant	Food and Preparation
Animal and By-product	Vegetable, Fruit and Plant	Food and Preparation	Mineral and Stone
Vegetable, Fruit and Plant	Food and Preparation	Mineral and Stone	Chemicals
Food and Preparation	Mineral and Stone	Chemicals	Raw Manufacturing
Mineral and Stone	Chemicals	Raw Manufacturing	Wood Product
Raw Manufacturing	Raw Manufacturing	Wood Product	Clothing and Accessory
Wood Product	Wood Product	Clothing and Accessory	Textile
Textile	Textile	Textile	Metal
Metal	Metal	Metal	Transportation
Transportation	Transportation	Transportation	Weapon
Weapon	Weapon	Weapon	Precious Stone

Iteration 12	Iteration 13	Iteration 14	Iteration 15
Daily Instrument	Raw Manufacturing	Service	Animal and By-product
Service	Service	Animal and By-product	Vegetable, Fruit and Plant
Animal and By-product	Animal and By-product	Vegetable, Fruit and Plant	Food and Preparation
Vegetable, Fruit and Plant	Vegetable, Fruit and Plant	Food and Preparation	Mineral and Stone
Food and Preparation	Food and Preparation	Mineral and Stone	Chemicals
Mineral and Stone	Mineral and Stone	Chemicals	Chemical Manufacturing
Chemicals	Chemicals	Chemical Manufacturing	Raw Manufacturing
Chemical Manufacturing	Chemical Manufacturing	Raw Manufacturing	Wood Product
Raw Manufacturing	Wood Product	Wood Product	Clothing and Accessory
Wood Product	Clothing and Accessory	Clothing and Accessory	Textile
Clothing and Accessory	Textile	Textile	Metal
Textile	Metal	Metal	Transportation
Metal	Transportation	Transportation	Weapon
Transportation	Weapon	Weapon	Daily Instrument
Weapon	Daily Instrument	Daily Instrument	Precious Stone
Precious Stone	Precious Stone	Precious Stone	Service
High-tech	High-tech	High-tech	High-tech
Machinery	Machinery	Machinery	Machinery

Table 7: Marginal Opportunity Gain in an Iterative Process

Note that high-tech and machinery show negative marginal opportunity gain at the final stable stage, but that does not mean the two industries are valueless for the development Brazil. Instead, the suggestion means that Brazil, given to its economic situation at the final stage, is encouraged to prioritize the other industries and develop those bottom two industries only after it builds a balanced economic structure in the first place. The industries with negative opportunity gains, though not prioritized in their particular stage, should therefore still be proportionately strengthened within the capability of the country and not left behind.

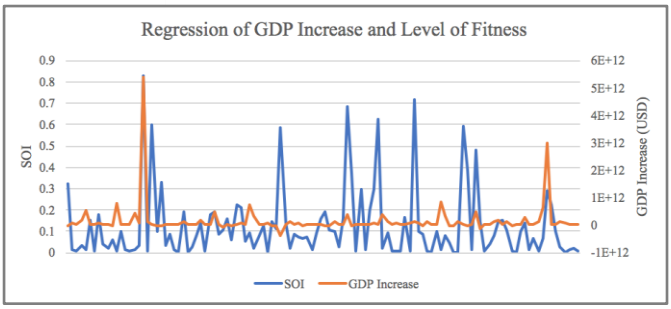
5. Macro-level Predictions

5.1 Structural Optimality Index

Indeed, such choice of prioritization has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our developmental strategies are and how deviations from the suggested developmental path will lead to non-optimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the R-Square obtained from regressing the country's actual development over the suggested development strategy. Both strategies are quantified in nature, with the actual development represented by export increase in major industry groups and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The R-square from the regression thus provides us with a concrete measure of how the development path adopted by the country has fitted with/differed from our suggestion.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which as defined as the net increase in GDP (USD), over their SOI. Figure 6 illustrates the regression output, which shows a strong positive correlation between countries' level of fitness and their five-year GDP growth, significant at the 1% level and achieving an adjusted R^2 of 11.9%. In other words, the more a country develops as suggested by calculations on a complexity level, the better economic outcomes it achieves.

Statistics	Value



Multiple R	0.355
R-Square	0.126
Adjusted R^2	0.119
Variable	**Value**
Coefficient	$1.249*10^{12}$
t-stat	4.061
P-value	0.000(***)

Figure 6: Regression of Five-year GDP Growth over SOI

5.2 SOI of Brazil and Its Implications

Brazil, for instance, has a particularly low level of structural optimality as compared with its emerging peers. As the regression in Figure 7 suggests (the orange curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), Brazil demonstrates a lowlevel of correlation with the suggested developmental path, having an adjusted R-square of merely around 0.8%. Brazil ranks the 99thplace among all countries in terms of SOI and has the slowest economic progress among its emerging peers from 2009 to 2014, which is consistent with the regression result in Figure 6. This is in direct contrast to emerging economies such as China, where there is a highSOI of around82.8% and a corresponding GDP growth of 5.39 billion USD, making it one of the fastest developing economies during the period. In sum, Brazil's stagnant economic progress from 2009 to 2014 can be at least partially explained by a mismatched development strategy that demonstrates a considerable level of deviationsfromour suggested developmental path. In specific, Brazil invested heavily in the service industry, while it should have focused more on basic industry groups such as animal and by-product.

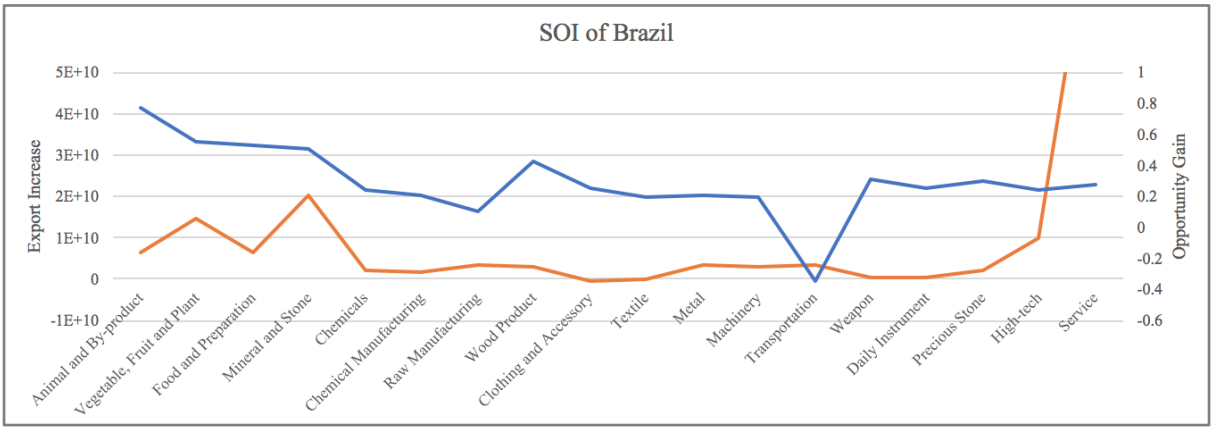


Figure 7: SOI ofBrazil (2009-2014)

6. Concluding Remarks

In conclusion, Brazil demonstrated a moderate level of economic growth with an annual GDP growth rate of 8.1%, concurrent with a decrease in economic complexity as compared with other major emerging economies during the period from 2009 to 2014.

The level of economic progress during the period, we believe, can be partially explained through analysis of the economic and product complexity structure of Brazil on the industry group level. The product classification system we adopt allow us to measure the actual development for various industry groups and to observe that Brazil aggressively invested in developing daily instrument, high-tech and machinery industries during the period, while its emerging peers and developed economies around the globe have adopted paths that are more well-suited to their current stances in the product space.

Based on our analysis of the opportunity gain and a sensitivity analysis of major industry groups, Brazil could further its gains by prioritizing stone, service, and clothing industry groups. We further note that Brazil has an ill-matched development strategy that indicates a particularly low level of structural optimality on the complexity level, having an adjusted R-square of merely 0.8%. Since a significant positive correlation exists between five-year GDP growth and the structural optimality index of an economy as we define and calculate, Brazil indeed has a suboptimal growth experience as compared with fast-growing economies both on a regional scale and on a global scale, and has the potential to further its economic gains through industrial structure optimization as suggested in the paper.